

Cross-Linked Polymer Synthesis

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Synonyms

[Chemical cross-linking](#), [Cross-linking between polymer chains](#), [Cross-linking during polymerization](#), [Photo-curing](#), [Physical cross-linking](#), [Sol-gel transition](#), [Thermal curing](#)

Definition

Polymers can be cross-linked by chemical bonding or physical interaction. Synthetic methods for chemically cross-linked polymers are classified into cross-linking during polymerization and post-cross-linking of polymer chains. In the former method, chain-growth polymerization like radical polymerization as well as step-growth polymerization like polycondensation and polyaddition can be used for the construction of cross-linked polymers. Chain-growth polymerization of a monomer with one polymerizable group (e.g., styrene) in the presence of that with two or more polymerizable functional groups (e.g., divinylbenzene) gives the corresponding cross-linked polymer. On the other hand, the formation of cross-linked polymers in step-growth polymerization can be accomplished by using a monomer with three or more functional groups as a component of monomers. Post-cross-linking between polymer chains is usually carried out by the reaction between reactive groups on the polymer chains and a cross-linker having two or more reactive groups. Polymerization of functional groups introduced onto polymer chains also gives the cross-linked polymers. Cross-linking reactions are usually promoted by heating or photoirradiation, though the addition of catalysts and the irradiation of radioactive rays are also used for promotion of cross-linking.

Physical cross-linking is performed using interactions other than the covalent bond, such as hydrogen bonding or ionic interaction. Physical cross-links can be reversibly dissociated and recombined under specific stimuli such as heating/cooling. Physical de- and re-cross-linking in the presence of a solvent are observed as sol–gel transition. Novel kinds of cross-links such as sliding structures and dynamic covalent bonds are recently utilized for the formation of cross-linked polymers.

Introduction

When a number of linear polymer chains are interconnected at several points on the chains, they finally become a single macromolecule with a network structure. This kind of interconnection is called cross-linking, and the finally obtained polymer is called a network polymer. The interconnected points are called cross-linking points or cross-links. Cross-links are categorized as covalently bonded chemical cross-links or physical cross-links. Physical cross-links interconnect

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polymer chains using interactions other than covalent bonding, such as hydrogen bonding, hydrophobic interaction, ionic bonding, and coordination bonding.

In contrast to linear polymers, cross-linked polymers are insoluble in any solvents, though they can swell in good solvents for the corresponding linear polymers. Polymers cross-linked by covalent bonds or strong physical interactions do not melt and finally decompose when they are heated. Based on these properties and depending on their degree of cross-linking, cross-linked polymers are used mainly for three types of applications. Polymers with a small amount of cross-links and a good swelling property are utilized as gel materials. Gels can absorb and preserve a large amount of solvent, and therefore, they are applied to sanitary napkins, disposable diapers, soft contact lenses, water retention materials, and so on. Vulcanization of diene polymers, such as poly(1,3-butadiene) and polyisoprene, gives polymers cross-linked by a small amount of sulfur linkages and they are widely used as rubbers. On the other hand, chemically cross-linked polymers with high cross-link density hardly swell in solvents and can be used as materials with good thermal and mechanical properties and high chemical resistance. Representative examples of such polymers are thermosetting resins. They are cross-linked by heating and the resulting highly cross-linked network polymers are utilized as constructional, electric, and electronic materials; adhesives; and so on.

Physical cross-links effectively work in biological gels such as bacterial cellulose and crystalline lens and vitreous body in the eyeball. They are also used in gels prepared from natural polymers like agar and gelatin. In addition, physical cross-links play an important role in synthetic polymers such as thermoplastic elastomers and thixotropic paints.

Syntheses of Chemically Cross-Linked Polymers

Synthetic methods for chemically cross-linked polymers are categorized as (1) formation of a network structure from monomers by polymerization or (2) post-cross-linking of linear polymers with a cross-linking agent.

Formation of Cross-Linked Structure by Polymerization

In self-polycondensation of multifunctional monomer (the number of functional groups = f), if there is no intramolecular condensation, the following equations hold between the number- and weight-average degrees of polymerization (X_n and X_w , respectively) and the degree of reaction of the functional groups (p):

$$X_n = 1/(1 - (fp/2)) \quad (1)$$

$$X_w = (1 + p)/\{1 - (f - 1)p\} \quad (2)$$

From these equations, the degrees of reaction of the functional groups when X_n and X_w reach infinite values (p_n and p_w , respectively) are written as

$$p_n = 2/f \quad (3)$$

$$p_w = 1/(f - 1) \quad (4)$$

When $f \geq 3$, p_w becomes smaller than p_n . Therefore, the degree of reaction at which a three-dimensional network structure is constructed (gel point (p_{gel})) is described as

$$p_{gel} = p_w = 1/(f - 1) \quad (5)$$

For example, when a monomer with two functional groups is reacted with that with three functional groups, where the functional groups of each monomer are fed in equivalent, the average f value is $2 \times 3/5 + 3 \times 2/5 = 2.4$, and therefore, $p_{\text{gel}} = p_w = 0.71$. This means that X_w diverges and gelation starts when 71 % of the functional groups react.

On the other hand, in chain-growth copolymerization between a monofunctional monomer (such as vinyl monomers) and a difunctional monomer (such as divinyl compounds), cross-linking progresses together with polymerization and finally the gel is formed. In this case, if all functional groups have equal reactivity and there is no intramolecular cyclization, the degree of reaction of the functional groups at the gel point is written as

$$p_{\text{gel}} = 1/\{(x_w - 1)\rho\} \quad (6)$$

In Eq. 6, x_w is the weight-average degree of polymerization of primary polymer chains, and ρ is the ratio of the functional groups belonging to the difunctional monomer to all those in the system.

Cross-linking using step-growth polymerization has been widely utilized as curing reactions of thermosetting resins. The representative examples are polycondensation curing reactions of phenol-formaldehyde resins (novolac and resol), urea-formaldehyde resins, and melamine-formaldehyde resins. Another example is a polyaddition curing reaction of epoxy resins, which consist of the compounds having two or more epoxy groups (oxirane rings). A wide variety of epoxy resins with various molecular structures and molecular weights have been developed, and a broad range of properties can be added to cured epoxy resins by changing curing agents and/or additives. This versatility enables the use of epoxy resins to various applications.

Typical epoxy resins are glycidyl ethers, glycidyl esters, glycidyl amines, and cycloaliphatic epoxides that are synthesized by oxidation of the corresponding olefins by peracid or hydrogen peroxide (Fig. 1). Glycidyl ethers are the most widely used epoxy resins, and especially, diglycidyl ether of bisphenol A is the most popular.

In curing of epoxy resins by polyaddition, polyfunctional primary or secondary amines with aliphatic or aromatic backbone structures are often used as curing agents. Other representative curing agents are acid anhydrides, polyfunctional phenols, and phenolic resins. Curing of epoxy resins can also be accomplished by ring-opening polymerization of epoxy groups by using acids or tertiary amines as initiators. These curing reactions are shown in Fig. 2.

To obtain cured thermosetting resins with high thermal and mechanical properties, highly developed three-dimensional network structures should be prepared by accomplishing sufficient progress of cross-linking reactions during the curing process. Curing of a resin at a single temperature is not often efficient for it because cross-linking reactions are terminated before construction of sufficient network structures. The glass transition temperature (T_g) of the resin is increased as the progress of the cross-linking reaction and finally exceeds the curing temperature, making the resin glassy and inhibiting the micro-Brownian motion of the resin segments for further cross-linking reaction. To prevent this insufficient cross-linking, multistep curing by increasing curing temperatures at regular time intervals or high-temperature post-curing should be carried out.

Appropriate selection of molecular structures, curing agents, and curing conditions enables facile production of cured resins having good thermal property with T_g of more than 200 °C. Thermoplastic polymers with similar thermostability can be obtained by proper molecular design such as the introduction of rigid structure, but their processing, which requires heating above melting temperature, is more difficult than that of thermosetting resins.

Chain-growth polymerization is usually faster than step-growth polymerization, inducing faster chain propagation and cross-linking, and the polymer chain ends (radicals, anions, or cations)

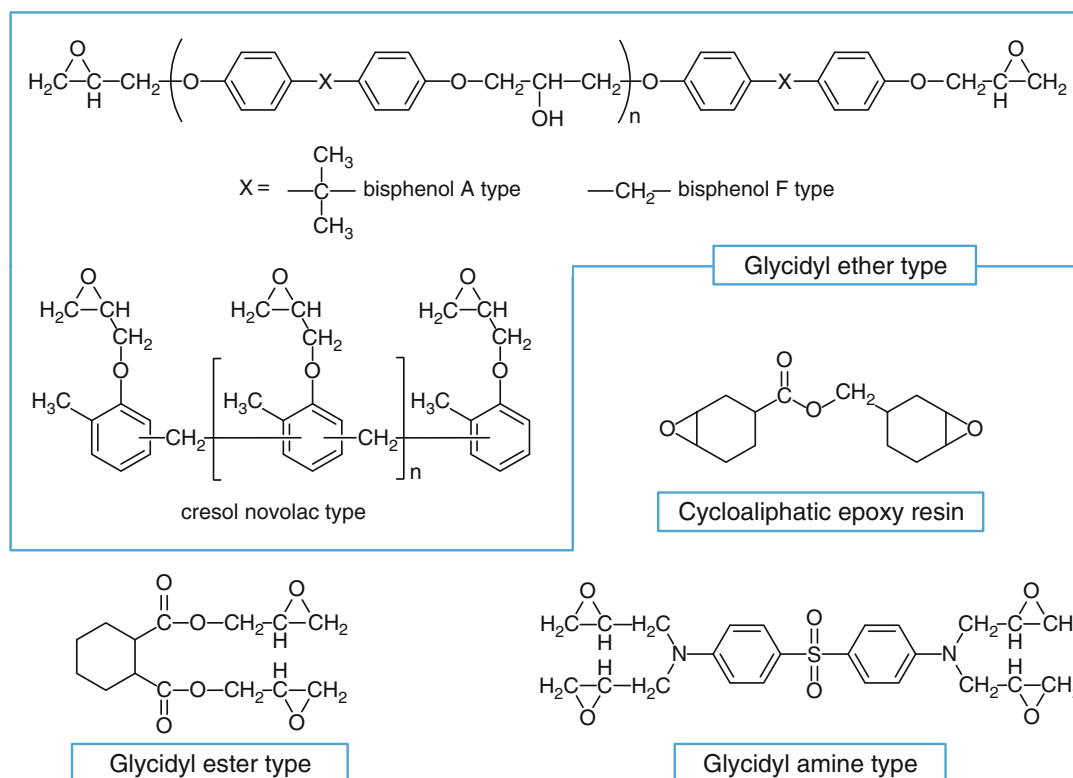


Fig. 1 Structures of typical epoxy resins

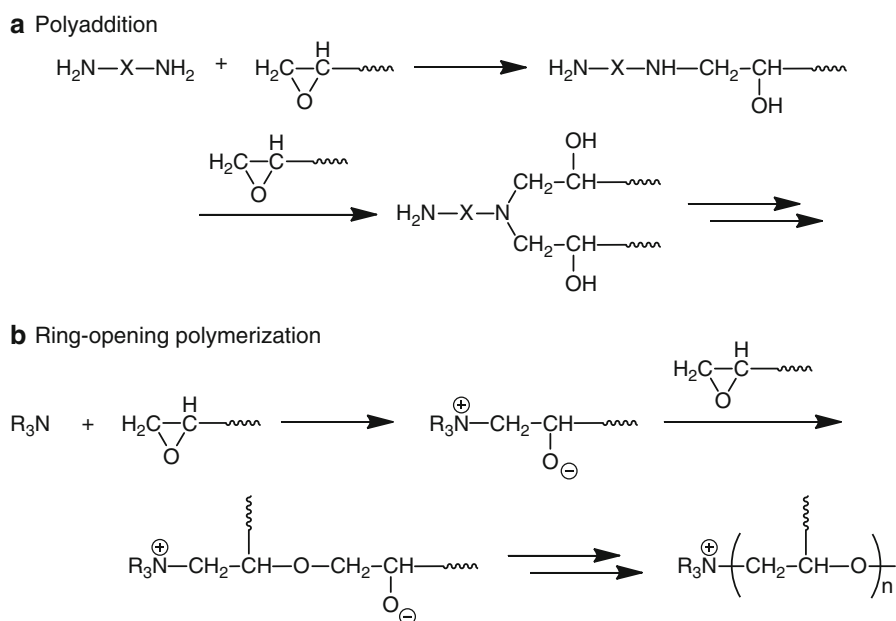


Fig. 2 Examples of curing reactions of epoxy resins

readily lose the activity by termination or chain-transfer reactions. Thus, therein, multistep curing usually does not improve the degree of cross-linking. Cross-linking by chain-growth polymerization is often carried out by using a large amount of monofunctional monomer with a small amount of

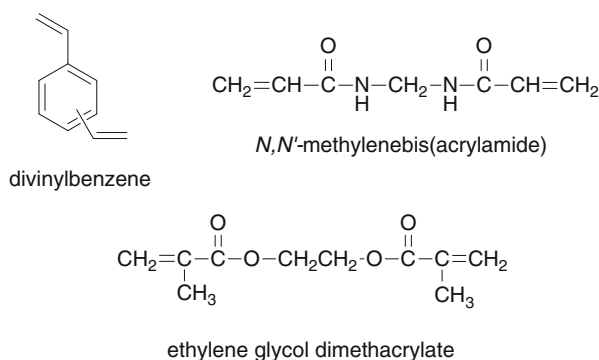


Fig. 3 Examples of cross-linking agents for vinyl polymerization

difunctional monomer (cross-linking agent) in the presence of a solvent, and in this case, the resultant gel is used without further purification. Gel particles prepared by suspension copolymerization between styrene, chloromethylstyrene, and divinylbenzene have been widely used for solid-phase synthesis of polypeptides (Merrifield resin).

Radical polymerization, which is most widely used for cross-linking in chain-growth polymerization, is usually initiated by heating or photoirradiation in the presence of thermo- or photoinitiators, respectively. Therein, various vinyl monomers can be used as monofunctional monomers. Some monomers such as *N*-isopropylacrylamide and *N*-vinylisobutylamide produce characteristic gels, which have a lower critical solution temperature (LCST) property, swelling at a low temperature and shrinking at a high temperature. Monomers with two or more vinyl groups can be used as cross-linking agents; divinyl compounds are most commonly used. Using a cross-linking agent with similar structure and reactivity to the copolymerized vinyl monomers is desirable to obtain a polymer with a homogeneous network structure. Divinylbenzene, *N,N'*-methylenebis(acrylamide), and ethylene glycol dimethacrylate are typical examples of cross-linking agents (Fig. 3). Recently, attempts to obtain gels with very uniform molecular weight and molecular weight distribution between cross-linking points have been carried out by use of living radical polymerizations [1, 2].

Cross-Linking Between Polymer Chains

Intermolecular reactions between linear polymers with or without cross-linking agents, which have two or more functional groups to react with the polymers, result in the formation of cross-linked network polymers. In principle, any kind of coupling reactions can be used for the interpolymer cross-linking. Typical examples are the formation of sulfide linkages by vulcanization of diene polymers with sulfur, cross-linking of poly(vinyl cinnamate) by photodimerization, and acetalization between poly(vinyl alcohol) and difunctional aldehydes (Fig. 4). Cross-linking by polymer reactions between linear polymers facilitates the formation of a homogeneous network structure and the control of degree of cross-linking. On the other hand, cross-linking by chain-growth copolymerization between a vinyl monomer and a divinyl compound tends to form an inhomogeneous network structure due to the very fast polymerization rate and the difference in reactivity between the monomer and the cross-linking agent. In cross-linking based on a polymer reaction, the concentration of the reaction system is important, and the reaction under low concentration often results in insufficient interpolymer cross-linking because of the preferential progress of intramolecular reactions induced by the so-called polymer effect.

As in the case of usual organic reactions, cross-linking reactions are usually promoted by heating or photoirradiation, as well as by the addition of catalysts and the irradiation of radioactive rays.

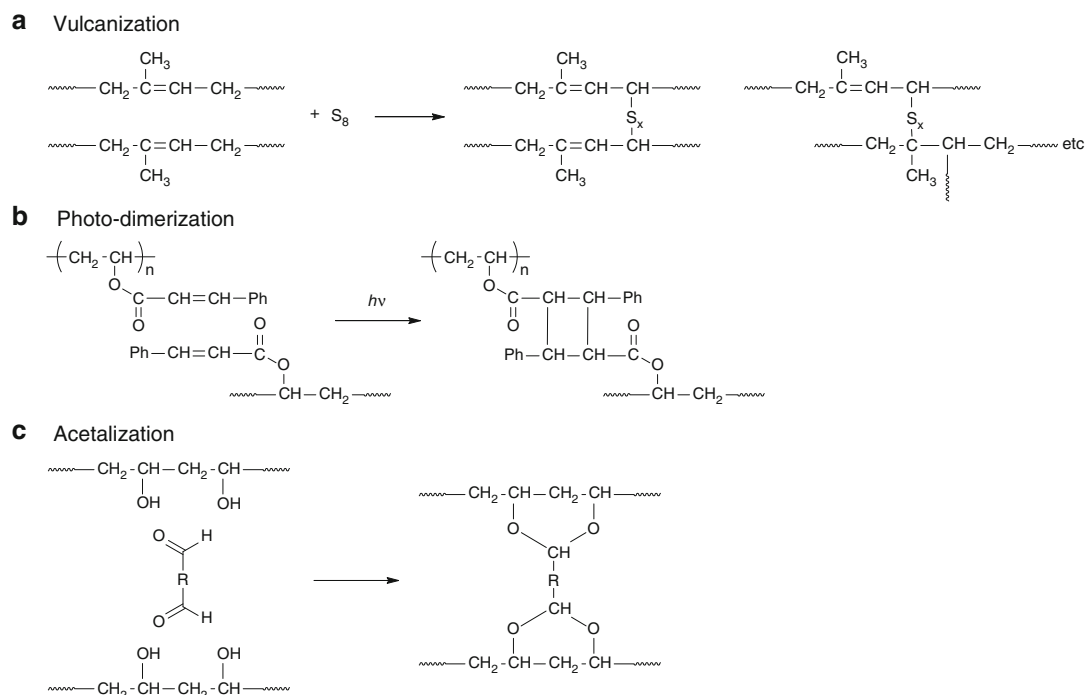


Fig. 4 Examples of interpolymer cross-linking reactions

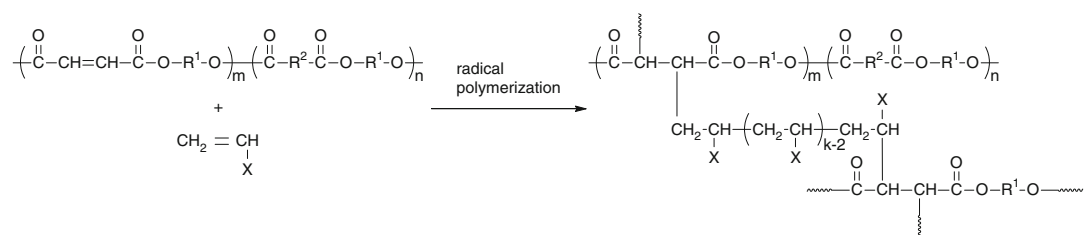


Fig. 5 Cross-linking reaction of unsaturated polyester resins

The click reaction between azide and alkyne in the presence of copper(I) catalyst (the Huisgen reaction) has been used for cross-linking of biopolymers due to its progress under mild conditions [3]. When a gamma ray is irradiated to an aqueous solution of poly(vinyl alcohol), hydroxyl radicals generated from water abstract hydrogen atoms on the main chain of poly(vinyl alcohol) and the resulting polymer radicals are coupled with each other to give a cross-linked structure [4].

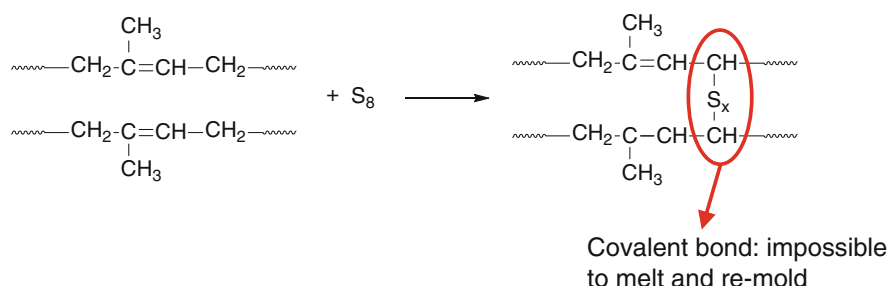
Polymerization reactions can also be used for cross-linking reactions between polymer chains. In the curing reaction of unsaturated polyester resins, radical copolymerization between a vinyl monomer (e.g., styrene) and double bonds of unsaturated polyesters, which are prepared by polycondensation of unsaturated and saturated dicarboxylic acids with a glycol, is used as the cross-linking reaction (Fig. 5). Unsaturated polyester resins are used as matrices for fiber-reinforced plastics (FRPs) containing inorganic fibers, such as glass fibers and carbon fibers, as reinforcing materials. FRPs based on unsaturated polyester resins have been applied to various uses including bathtubs, tanks, construction materials such as corrugated and flat plates, pipes, boats, and automobiles.

Synthesis of Physically Cross-Linked Polymers

Physical cross-links are defined as those using interactions other than the covalent bond, such as coordination bonding, hydrogen bonding, ionic interaction, or van der Waals interaction. Though chemical cross-linking forms covalent bonds and reversible dissociation of the prepared cross-links is impossible, physical cross-links can be, at least in principle, reversibly dissociated and recombined. This difference between chemical and physical cross-links is best illustrated by contrasting commodity rubbers, which are produced by vulcanization, with thermoplastic elastomers utilizing block copolymers. Vulcanization with a cross-linking agent (e.g., sulfur) forms covalently bonded cross-links between polymer chains, and therefore, it is impossible to melt and remold a rubber once molded and vulcanized (Fig. 6a). In contrast, thermoplastic elastomers are block copolymers having hard segments that do not flow at room temperature due to interactions between the segments, such as hydrogen bonding, ionic interaction, crystallization, and van der Waals interaction, and also having soft segments with a low glass transition temperature. These two segments exhibit microscopic phase separations, and as a result of physical cross-links at the hard segments, thermoplastic elastomers express rubbery properties (Fig. 6b). Therefore, they can be remelted and remolded through dissociating physical cross-links by heating.

Gelation of an aqueous solution by physical cross-linking based on hydrogen bonds is observed when agar or gelatin is dissolved by heating and then cooled. Addition of Ca^{2+} to an aqueous solution of alginate affords a gel with physical cross-links based on ionic interactions. Formation of

a Vulcanized rubber



b Thermoplastic elastomer

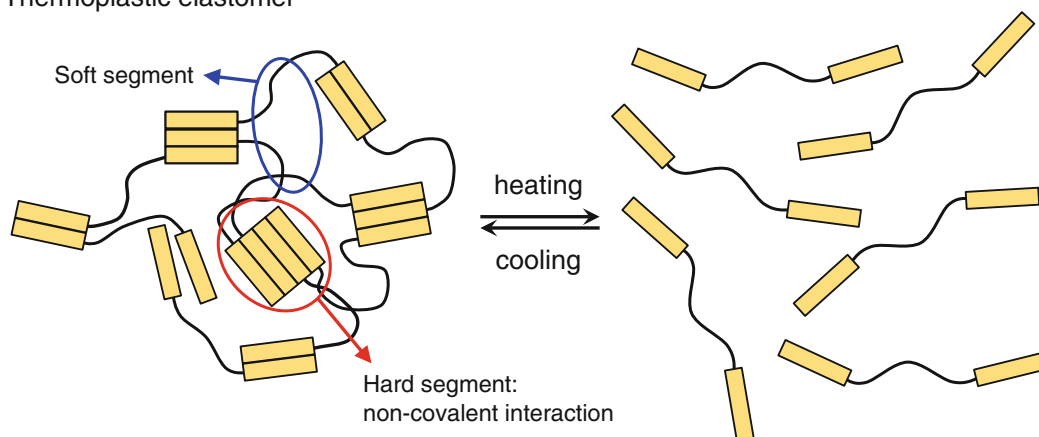


Fig. 6 Difference between chemically and physically cross-linked elastomers

a polyion complex between polycation and polyanion (e.g., poly(vinylbenzyltrimethylammonium chloride) and poly(sodium styrenesulfonate)) also gives a gel based on physical cross-links.

Use of Novel Cross-Link Structures

Recently, various cross-links that cannot be categorized as conventional chemical or physical cross-links have been utilized for the formation of network structures. Interpenetrating polymer networks (IPNs) are materials composed of two or more chemically cross-linked network structures that are mutually entangled and physically inseparable (Fig. 7). Formation of IPNs enables homogeneous mixing of mutually immiscible polymers. Generation of the second flexible network in the first electrolyte gel, which is composed of harder and more brittle network, affords a double-network (DN) gel having an IPN structure (Fig. 8) [5]. When an external force is added to the DN gel, the brittle network is firstly fractured as sacrificial bonds, and the fractured clusters work as physical cross-links of the second flexible network. As a result, the second ductile polymer chains act as hidden length and dissipate fracture energy by extensive elongation, and therefore, DN gels show remarkably large fracture energy ($4,400 \text{ J/m}^2$) in spite of their high water content ($\sim 90 \text{ wt}\%$).

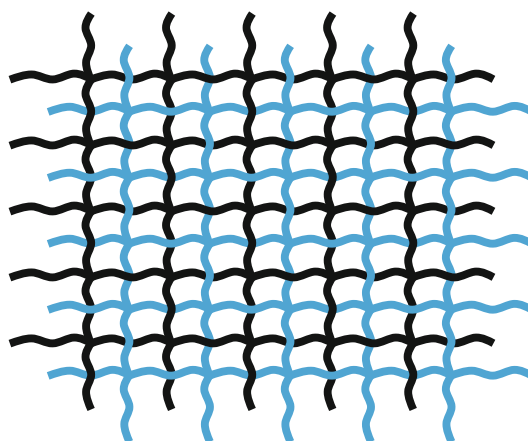


Fig. 7 Interpenetrating polymer network (IPN)

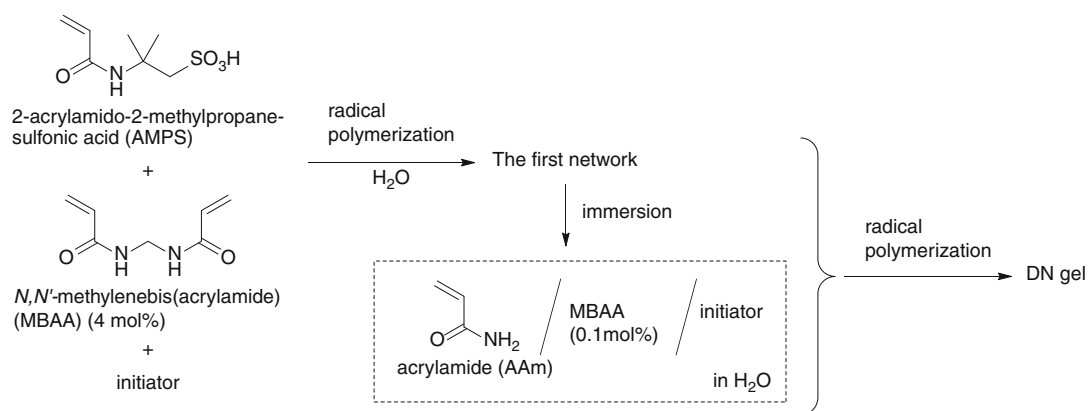


Fig. 8 Preparation of double-network (DN) gel

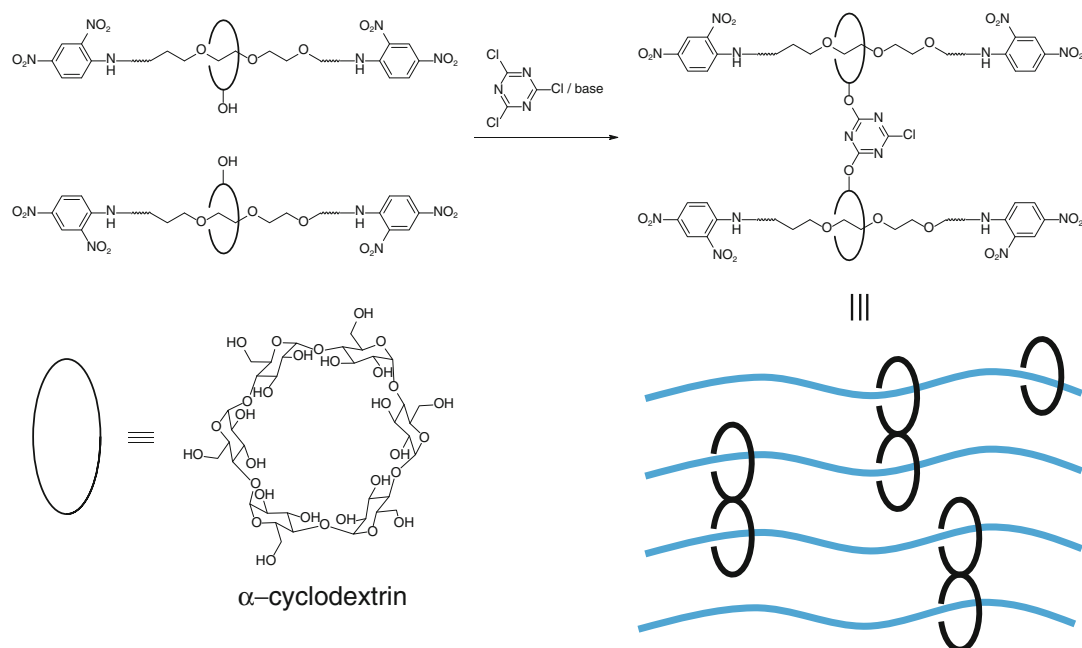


Fig. 9 Preparation of slide-ring gel

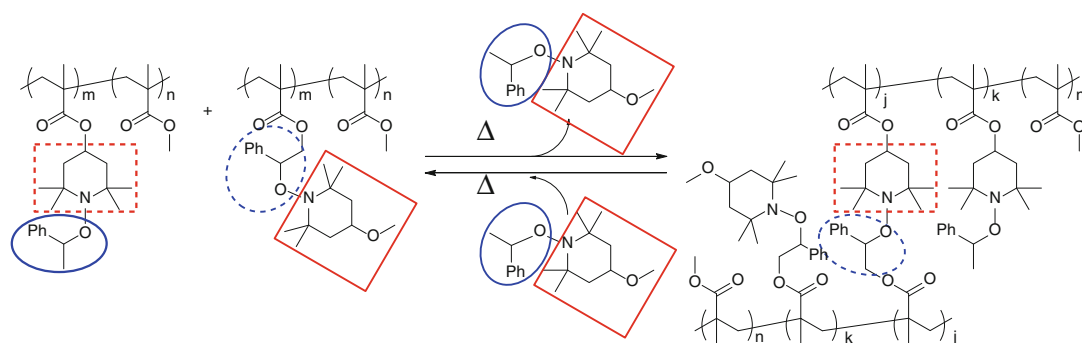


Fig. 10 Reversibly cross-linkable gel based on dynamic covalent chemistry

Topological gels using supramolecular rotaxane structures as cross-linking points have also been reported. α -Cyclodextrin as a cyclic oligosaccharide is known to include poly(ethylene glycol) to give polyrotaxane [6], and by the connection between the α -cyclodextrins of different polyrotaxane supramolecules, a slide-ring gel containing the connected rotaxanes as cross-linking points is obtained (Fig. 9) [6, 7]. When an external force is added to the slide-ring gel, nonuniformity of the structure and the stress in the gel is dissipated by sliding of the rotaxane cross-linking points (a pulley effect). Thus, the slide-ring gel can be swelled up to 8,000 times its dry weight, and the gel containing 90 % water can be stretched to 14 times. Polymers having slide-ring cross-linking points have also been applied to coatings and elastomers.

Covalent bonds that can be reversibly cleaved and recombined under external stimuli such as heat and light are known as dynamic covalent bonds, and they have also been used as cross-links between polymer chains. Alkoxyamines are known to be reversibly dissociated to the corresponding carbon and nitroxyl radicals. By using side-chain interconversion of polymers having alkoxyamine groups as side chains, covalently cross-linked gels that can show reversible sol–gel transition have been realized (Fig. 10) [8, 9].

Nanocomposite hydrogels (NC gels) have also been prepared by free radical polymerization of *N*-isopropylacrylamide in the aqueous solution containing exfoliated and homogeneously dispersed clays [10]. In the NC gels, many high-molecular-weight poly(*N*-isopropylacrylamide) chains are grafted on the clay surface at one or both ends. As a result, the clays act as planar cross-linking points, and the NC gels show good transparency and excellent extensibility (more than 1,000 %) at high water content (more than 90 %).

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References

1. Zhang H (2013) Controlled/“living” radical precipitation polymerization: A versatile polymerization technique for advanced functional polymers *Eur Polym J* 49:579–600
2. Oh JK, Drumright R, Siegwart DJ, Matyjaszewski K (2008) The development of microgels/nanogels for drug delivery applications *Prog Polym Sci* 33:448–477
3. Wang H, Tang L, Tu C, Song Z, Yin Q, Yin L, Zhang Z, Cheng J (2013) Redox-responsive, core-cross-linked micelles capable of on-demand, concurrent drug release and structure disassembly *Biomacromolecules* 14:3706
4. Hatakeyama T, Yamauchi A, Hatakeyama H (1984) Studies on bound water in poly(vinyl alcohol) hydrogel by DSC and FT-NMR *Eur Polym J* 20:61–64
5. Wu ZL, Kurokawa T, Gong JP (2011) Novel developed systems and techniques based on double-network principle *Bull Chem Soc Jpn* 84:1295–1311
6. Harada A, Hashidzume A, Yamaguchi H, Takashima Y (2009) Polymeric rotaxanes *Chem Rev* 109:5974–6023
7. Kato K, Ito K (2011) Dynamic transition between rubber and sliding states attributed to slidable cross-links *Soft Matter* 7:8737–8740
8. Higaki Y, Otsuka H, Takahara A (2006) A thermodynamic polymer cross-linking system based on radically exchangeable covalent bonds *Macromolecules* 39:2121–2125

9. Maeda T, Otsuka H, Takahara A (2009) Dynamic covalent polymers: Reorganizable polymers with dynamic covalent bonds *Prog Polym Sci* 34:581–604
10. Haraguchi K, Takehisa T, Fan S (2002) Effects of clay content on the properties of nanocomposite hydrogels composed of poly(/Visopropylacrylamide) and clay *Macromolecules* 35:10162–10171